PNEUMONIA DETECTION SYSTEM

Deep Learning Project Evaluation

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
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1. Introduction

Pneumonia is a severe lung infection that affects millions of people worldwide and continues to be a major cause of morbidity and mortality, especially among children and elderly populations. Early detection and timely treatment are crucial for reducing complications and improving patient outcomes. Chest X-ray imaging is one of the most commonly used methods for diagnosing pneumonia, but manual interpretation of radiographs can be challenging, time-consuming, and highly dependent on expert radiologists. These challenges have created a growing need for automated and intelligent diagnostic systems that can support faster and more accurate disease detection.

With the advancement of artificial intelligence and deep learning, medical image analysis has emerged as a powerful tool in healthcare applications. In particular, Convolutional Neural Networks (CNNs) have shown significant success in extracting meaningful patterns from medical images and performing accurate disease classification. This project explores the application of deep learning techniques for automated pneumonia detection using chest X-ray images, aiming to build a reliable classification model capable of distinguishing between normal lungs and pneumonia-infected lungs.

The project is based on the Kaggle Chest X-Ray (Pneumonia) dataset, which provides labeled chest radiographs for model training and evaluation. Through preprocessing, augmentation, feature learning, and performance optimization, the proposed system seeks to improve diagnostic accuracy while reducing dependency on manual screening. Evaluation using metrics such as accuracy, precision, recall, and F1-score further helps assess the model's effectiveness and reliability.

This work highlights the integration of deep learning with medical imaging to address a critical healthcare problem. By developing an automated pneumonia classification system, the project demonstrates the potential of artificial intelligence in supporting computer-aided diagnosis, assisting clinicians in decision-making, and improving early screening, particularly in resource-constrained healthcare environments.

2. Literature Survey

This section reviews ten significant studies related to pneumonia detection using deep learning, covering foundational models, transfer learning, explainability, and imbalance handling strategies relevant to this project.

- **CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays**
Rajpurkar et al. [1] proposed CheXNet, a DenseNet-121 model trained on ChestX-ray14 for pneumonia detection. The model achieved an F1 score of 0.435, surpassing radiologists' average performance. This work established DenseNet as a strong benchmark for automated chest X-ray analysis.

- **Custom CNN for Pneumonia Detection on Kaggle Dataset**
Stephen et al. [2] developed a custom CNN using the Kaggle Chest X-ray dataset and achieved 93.73% accuracy with high sensitivity and specificity. The study showed that custom CNNs can perform effectively without pretrained models.

- **Ensemble CNN Model for Pneumonia Detection**
Chouhan et al. [3] proposed an ensemble of pretrained CNNs including AlexNet, GoogleNet, DenseNet-121, and ResNet-50. The model achieved 96.4% accuracy, demonstrating the effectiveness of ensemble learning for improved generalisation.

- **Explainable Pneumonia Classification Using ResNet50 and Grad-CAM**
Liang and Zheng [4] used ResNet50 with Grad-CAM for explainable pneumonia detection, achieving 95.3% accuracy and 0.982 AUC. Their work highlights the importance of visual interpretability in medical diagnosis.

- **Transfer Learning with SVM for Pneumonia Classification**
Varshni et al. [5] combined pretrained CNN feature extraction with SVM classification, where InceptionV3-SVM achieved 83% accuracy. This study provides insight into hybrid deep learning approaches.

- **VGG19 with CLAHE Preprocessing for Pneumonia Detection**

Nair et al. [6] applied CLAHE preprocessing with VGG19 and class-weighted loss, achieving 90.1% accuracy and high pneumonia sensitivity. The work demonstrates the benefit of preprocessing and imbalance handling.

- **DenseNet with Focal Loss and SMOTE**

Jaiswal et al. [7] addressed class imbalance using DenseNet-121 with focal loss and SMOTE, achieving an AUC of 0.96. Their approach shows the impact of imbalance mitigation strategies on performance.

- **ChestX-ray14 and Weakly Supervised Localization**

Wang et al. [8] introduced ChestX-ray14 and used DenseNet-121 with class activation maps for disease localization. The study laid the foundation for explainability methods such as Grad-CAM.

- **Hybrid CNN–Machine Learning Ensemble**

Mittal et al. [9] combined CNN feature extraction with classifiers like Random Forest and SVM, achieving 94.8% accuracy. This work provides a comparative baseline for hybrid classification pipelines.

- **CheXpert: Large-Scale Chest Radiograph Dataset**

Irvin et al. [10] introduced CheXpert and explored uncertainty label handling strategies using DenseNet-121. The study demonstrated strong performance and provides valuable insight for extending pneumonia detection to broader diagnostic settings.

Table 1: Comparative Summary of Reviewed Studies on Pneumonia Detection from Chest X-Rays

References	Dataset	Methodology	Key Results
Rajpurkar et al., 2017 [1]	ChestX-ray14 (112,120 images, 14 diseases)	121-layer DenseNet trained end-to-end with binary cross-entropy loss for pneumonia classification	F1 score 0.435 on pneumonia class, surpassing mean radiologist F1 of 0.387
Stephen et al., 2019 [2]	Kaggle Chest X-ray (5,863 images)	Custom CNN with 4 convolutional blocks, max-pooling, dropout, dense layers, ReLU activations, and binary cross-entropy loss	Accuracy 93.73%, sensitivity 95%, specificity 93.3%
Chouhan et al., 2020 [3]	Kaggle Chest X-ray (5,863 images)	Ensemble of AlexNet, GoogleNet, DenseNet-121, and ResNet-50 fine-tuned with data augmentation and weighted majority voting	Accuracy 96.4%, F1 score 96.3%; ensemble significantly outperformed individual models
Liang & Zheng, 2020 [4]	Kaggle Chest X-ray + external validation sets	ResNet50 fine-tuned for binary classification with Grad-CAM visualisation for explainability	Accuracy 95.3%, AUC 0.982; Grad-CAM highlighted clinically relevant opacity regions
Varshni et al., 2019 [5]	Kaggle Chest X-ray (5,863 images)	Pretrained VGG16, InceptionV3, ResNet50 as frozen feature extractors; penultimate-layer features fed to SVM with RBF kernel	InceptionV3 + SVM: 83% accuracy; hybrid CNN-ML outperformed softmax-only CNN
Nair et al., 2020 [6]	Kaggle Chest X-ray (5,863 images)	VGG19 fine-tuned with CLAHE preprocessing and class-weighted cross-entropy loss for class imbalance handling	Accuracy 90.1%, pneumonia sensitivity 97.5%; CLAHE improved feature contrast
Jaiswal et al., 2021 [7]	NIH + Kaggle Chest X-ray datasets	DenseNet-121 with focal loss and SMOTE oversampling to address severe class imbalance	AUC 0.96 for pneumonia; significantly outperformed standard binary cross-entropy training
Wang et al., 2017 [8]	ChestX-ray14 (108,948 images, 14 diseases)	Multi-label DenseNet-121 with global average pooling-based Class Activation Maps (CAMs) for weakly supervised localisation	Pneumonia AUC 0.768; interpretable CAM heatmaps identifying affected lung regions
Mittal et al., 2018 [9]	Kaggle Chest X-ray (5,863 images)	CNN feature extractor combined with Random Forest and SVM ensemble classifiers instead of softmax layer	Accuracy 94.8%; traditional ensemble classifiers effectively leveraged CNN-generated features
Irvin et al., 2019 [10]	CheXpert (224,316 images, 65,240 patients)	DenseNet-121 with five uncertainty label strategies (U-Ignore, U-Zeros, U-Ones, U-SelfTrained, U-MultiTask) evaluated against radiologists	AUC 0.930 for pleural effusion; competitive with radiologists across five pathologies

3. Objectives

I. To develop an automated deep learning-based system for classifying chest X-ray images into Normal and Pneumonia categories.

II. To utilize the Kaggle Chest X-Ray (Pneumonia) dataset for training, validating, and testing the proposed classification model.

III. To apply image preprocessing and data augmentation techniques to improve image quality, increase dataset diversity, and enhance model generalization.

IV. To design and train a Convolutional Neural Network (CNN) model capable of extracting meaningful features from chest X-ray images for accurate pneumonia detection.

V. To evaluate the performance of the proposed model using metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis.

VI. To minimize false classifications, particularly false negatives, to support reliable and early detection of pneumonia cases.

VII. To leverage advanced pretrained architectures such as ResNet50 and MobileNetV2 through transfer learning, enabling efficient feature extraction and improved classification accuracy while maintaining computational efficiency.

4. Methodology

4.1 Workflow

4.1.1 Data Collection and Dataset Preparation

The Kaggle Chest X-Ray (Pneumonia) dataset consisting of **5,863 pediatric chest X-ray images** is used for this project, containing two classes: **Normal** and **Pneumonia**. The dataset is organized into training, validation, and testing sets to ensure proper distribution for model development and evaluation. This structured dataset division supports effective learning, hyperparameter tuning, and unbiased model assessment.

4.1.2 Preprocessing and Data Augmentation

The chest X-ray images are preprocessed to improve data quality through resizing, normalization, and enhancement techniques. To improve model generalization and reduce overfitting, data augmentation methods such as rotation, flipping, zooming, and shifting are applied. These techniques increase dataset diversity and help the model learn robust disease-related features.

4.1.3 Model Development

Multiple deep learning architectures are implemented for pneumonia detection, including a baseline Convolutional Neural Network (CNN) along with advanced models such as ResNet and MobileNet. The CNN model serves as a baseline for performance comparison, providing a reference point to evaluate the effectiveness of more complex architectures. ResNet leverages residual connections to address the vanishing gradient problem, enabling the training of deeper networks and improving feature learning.

In contrast, MobileNet uses depthwise separable convolutions to reduce computational complexity while maintaining performance. All these models are trained to classify chest X-ray images as either normal or pneumonia-affected, allowing for a comprehensive comparison of their performance in automated disease detection.

4.1.4 Feature Learning and Classification

The models automatically extract meaningful features from chest radiographs using convolutional and pooling layers. These features capture critical patterns such as lung opacities and infection-related abnormalities. The extracted features are then passed through fully connected layers for binary classification, distinguishing between healthy and pneumonia-infected lungs.

4.1.5 Performance Evaluation and Analysis

Model performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis. Additionally, training and validation loss curves are monitored to assess convergence behavior and detect overfitting. A comparative analysis is conducted between CNN, ResNet, and MobileNet models to determine the most effective architecture for pneumonia detection in chest X-ray images.

4.2 Architecture

4.2.1 Base CNN Model

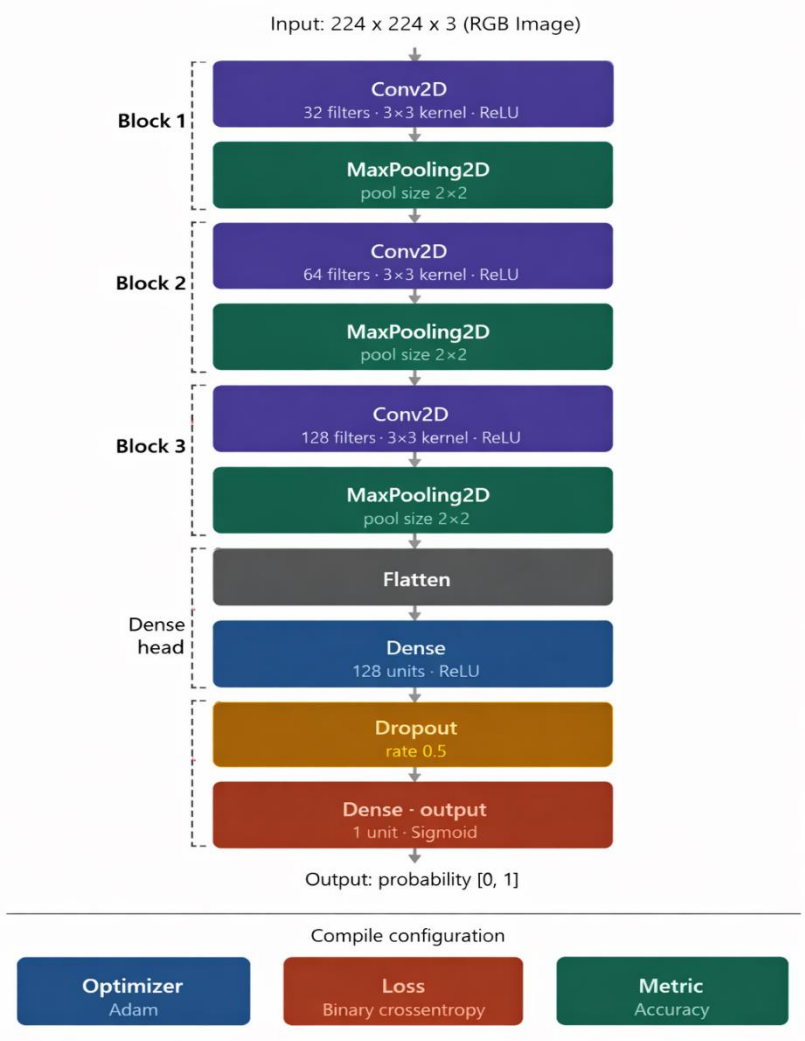

Fig 1. Sequential CNN architecture for binary image classification, showing convolutional, pooling, dense, and output layers with compile configuration.

Fig. 1 presents a Sequential Convolutional Neural Network (CNN) tailored for binary image classification, beginning with an input of 224 × 224 × 3 RGB images. The architecture is organized into three convolutional–pooling blocks: Block 1 applies a Conv2D layer with 32 filters of size 3×3 and ReLU activation, followed by a 2×2 max pooling layer; Block 2 expands to 64 filters with the same kernel and activation, again followed by 2×2 pooling; Block 3 further deepens the feature extraction with 128 filters and concludes with 2×2 pooling. The output of these blocks is flattened

and passed into the dense head, which consists of a fully connected layer with 128 units and ReLU activation, a dropout layer with a rate of 0.5 to mitigate overfitting, and a final dense output layer with a single unit and sigmoid activation that produces a probability between 0 and 1. The model is compiled using the Adam optimizer, binary cross-entropy loss, and accuracy as the evaluation metric, making **Fig. 1** a clear depiction of a standard CNN pipeline that progresses from low-level feature extraction to high-level decision making for two-class classification tasks.

4.2.2 ResNet50

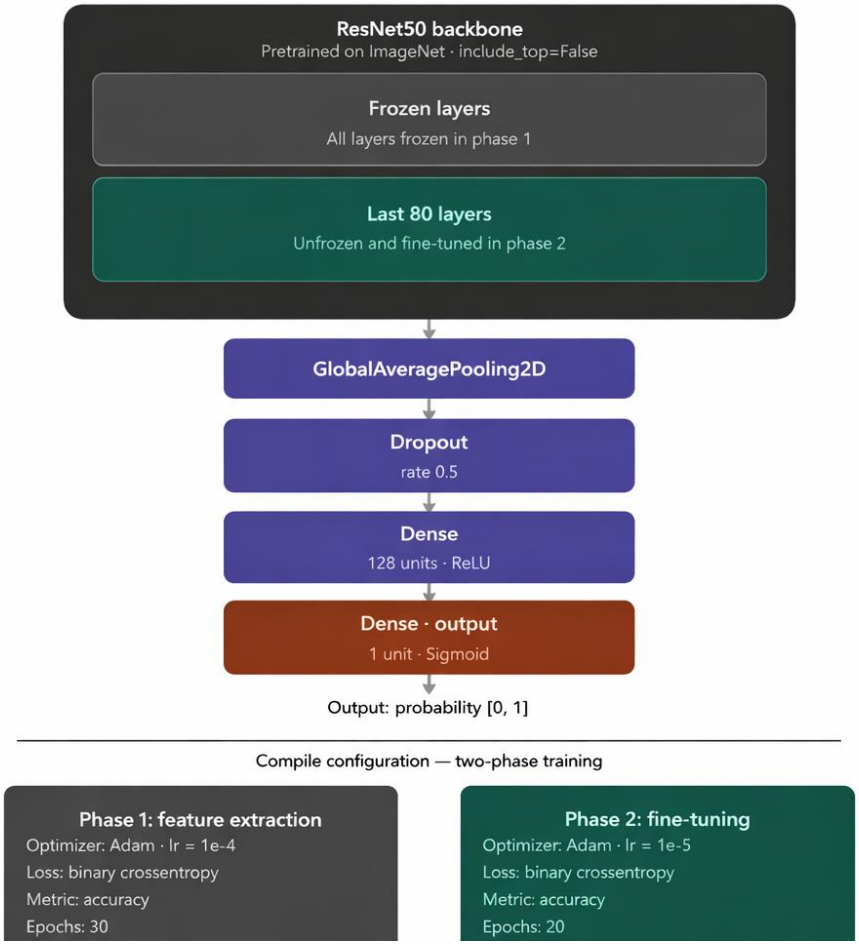

Fig. 2: Transfer learning workflow of a ResNet50 binary classifier showing two-phase training and fine-tuning process

Fig. 2 illustrates the complete transfer learning pipeline for a ResNet50-based binary classifier, emphasizing both architectural design and the two-phase training strategy. The model begins with an input of size $224 \times 224 \times 3$, which passes through the ResNet50 backbone pretrained on

ImageNet with include_top=False. In Phase 1, all backbone layers are frozen for feature extraction, ensuring pretrained weights remain intact while only the newly added layers—GlobalAveragePooling2D, Dropout (rate 0.5), and Dense (128 units, ReLU)—are trained. The final Dense output layer with Sigmoid activation produces a probability between 0 and 1 for binary classification. In Phase 2, the last 80 layers of ResNet50 are unfrozen for fine-tuning, allowing the model to adapt learned features more precisely to the target dataset. The diagram highlights compile configurations for both phases, showing different learning rates (1e−4 and 1e−5), optimizers, and callbacks such as learning rate scheduling.

4.2.3 MobileNetV2

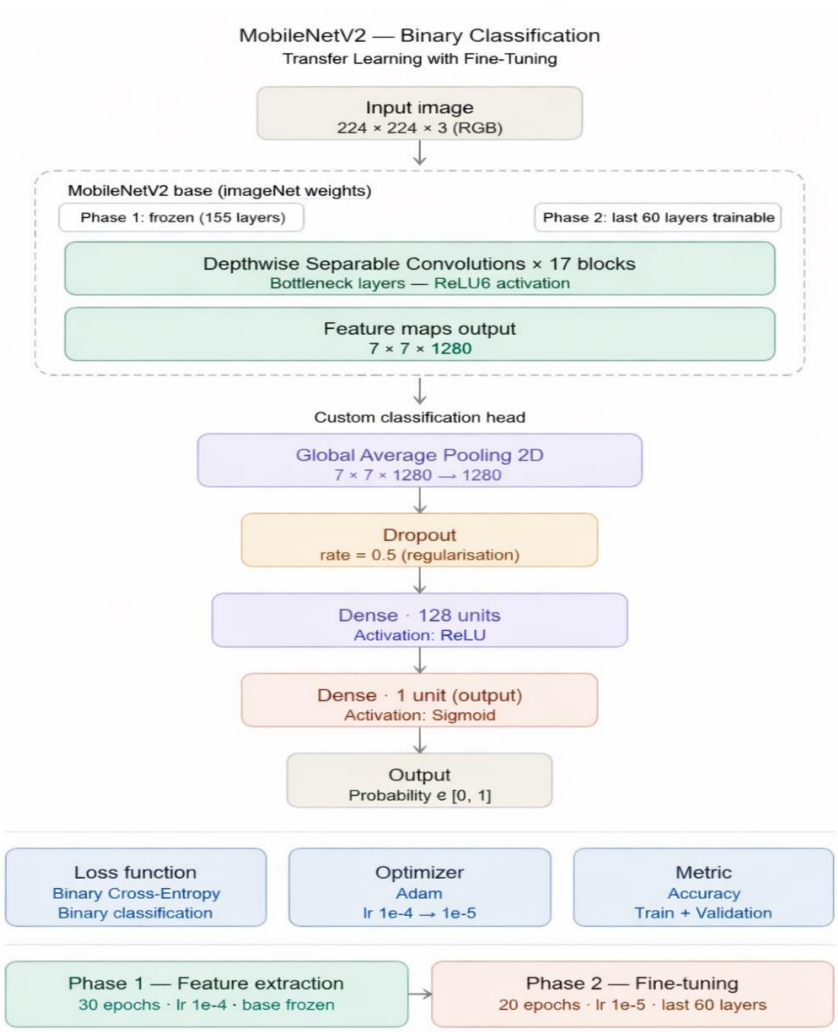

Fig. 3: Transfer learning workflow of a MobileNetV2-based binary classifier showing feature extraction and fine-tuning phases with corresponding training configurations.

Fig. 3 presents the full transfer learning pipeline for a MobileNetV2-based binary classifier, clearly outlining both the architectural components and the two-phase training strategy. The model begins with an input image of size 224 × 224 × 3, which is processed through the MobileNetV2 backbone pretrained on ImageNet. In Phase 1, the backbone layers are frozen to perform feature extraction, ensuring that the pretrained weights remain intact while only the custom classification head—comprising GlobalAveragePooling2D, Dropout (rate 0.5), and a Dense layer with 128 units and ReLU activation—is trained. The final Dense output layer with a Sigmoid activation produces a probability between 0 and 1 for binary classification. In Phase 2, the last 60 layers of MobileNetV2 are unfrozen for fine-tuning, allowing the model to adapt its learned features more precisely to the target dataset. The diagram also highlights compile configurations for both phases, including different learning rates (1e−4 and 1e−5), the Adam optimizer, and accuracy as the evaluation metric, thereby providing a comprehensive overview of how transfer learning leverages pretrained deep networks for efficient binary classification.

5. Implementation Details

The experiments were conducted using a consistent set of hyperparameters to ensure a fair comparison among the proposed Convolutional Neural Network (CNN), ResNet50, and MobileNetV2 architectures.

5.1 Input Configuration

All input images were resized to a fixed resolution of 224 × 224 × 3. Pixel values were normalized to the range [0, 1]. A batch size of **64** was used across training, validation, and testing datasets.

5.2 Data Augmentation

To improve generalization and reduce overfitting, data augmentation techniques were applied to the training dataset. These included random rotation (up to 20 degrees), zoom transformations (range of 0.2), and horizontal flipping. Validation and test datasets were only rescaled without augmentation.

5.3 Training Parameters

All models were trained using the Adam optimizer with an initial learning rate of 1×10^{-4}. During the fine-tuning phase, the learning rate was reduced to 1×10^{-5} to ensure stable convergence. The binary cross-entropy loss function was used due to the binary classification nature of the problem. Model performance was evaluated using accuracy as the primary metric.

5.4 Training Strategy

A two-stage transfer learning approach was adopted for deep learning models:

5.4.1 Feature Extraction Phase:

The convolutional base was frozen, and only the classification layers were trained for 30 epochs.

5.4.2 Fine-Tuning Phase:

The top layers of the base model (approximately last 60–80 layers) were unfrozen and trained jointly with the classification head for an additional 20 epochs using a reduced learning rate.

5.5 Regularization and Optimization

To improve model generalization, the following techniques were applied:

- Dropout: A dropout rate of 0.5 was used in the fully connected layer.
- Learning Rate Scheduling: A ReduceLROnPlateau strategy was employed with a reduction factor of 0.3, patience of 3 epochs, and a minimum learning rate of 1×10^{-6}.

5.6 Classification Head

The classification head consisted of a Global Average Pooling layer followed by a fully connected layer with 128 neurons (ReLU activation) and a final output layer with 1 neuron (Sigmoid activation).

5.7 Hardware Configuration

All experiments were conducted using Google Colab with access to an NVIDIA Tesla T4 GPU, enabling efficient training of deep learning models. The selected hyperparameters ensured stable training and allowed a fair and consistent comparison across all evaluated models.

6. Results and Analysis

The performance of the proposed models was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. A comparative analysis was performed between the custom CNN model and transfer learning-based models such as ResNet50 and MobileNetV2.

6.1 Quantitative Performance Evaluation

Table 2: Comparative Performance of all models.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
CNN	0.92	0.92	0.92	0.92
ResNet50	0.87	0.87	0.87	0.86
MobileNetV2	0.80	0.85	0.80	0.78

The CNN model achieved the highest accuracy of 92%, demonstrating superior performance compared to the other models.

6.2 Class-wise Performance Analysis

The CNN model achieved a recall of **0.99 for Pneumonia**, indicating excellent sensitivity and a strong ability to correctly identify infected cases, which is critical in medical diagnosis to minimize false negatives. ResNet50 demonstrated relatively balanced performance across both classes, with comparable recall values for Normal and Pneumonia. However, its overall performance was slightly lower, suggesting limited adaptation to domain-specific features. In contrast, MobileNetV2 exhibited a strong bias toward Pneumonia. While it achieved a high recall of **0.99** for Pneumonia, its recall for Normal cases was significantly lower (**0.49**), indicating frequent misclassification of Normal samples.

6.3 Loss Curve Analysis

The training and validation loss curves are shown in:

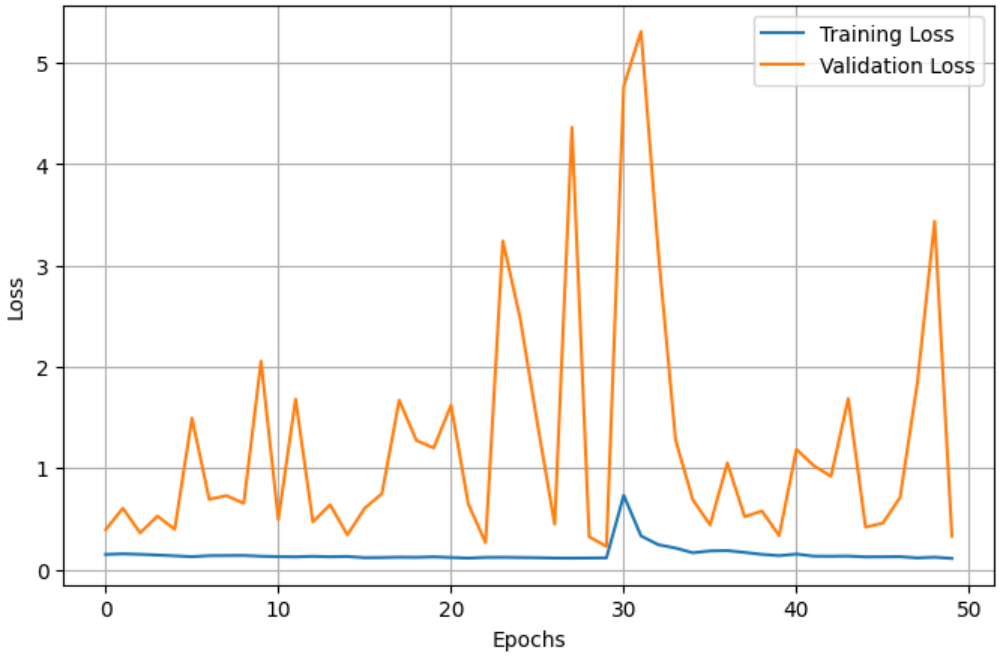

Fig. 4: ResNet50 Loss Curve

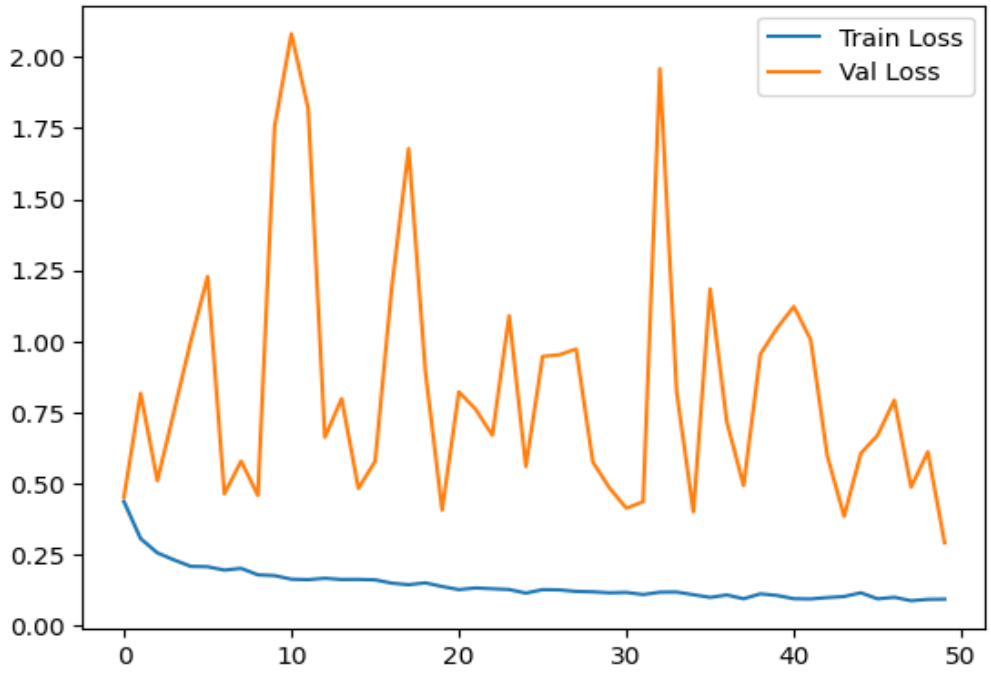

Fig. 5: CNN Loss Curve

The curves indicate effective learning with gradual convergence. The limited size of the validation dataset resulted in higher variance in validation loss, leading to fluctuations and minor overfitting.

6.4 Accuracy Curve Analysis

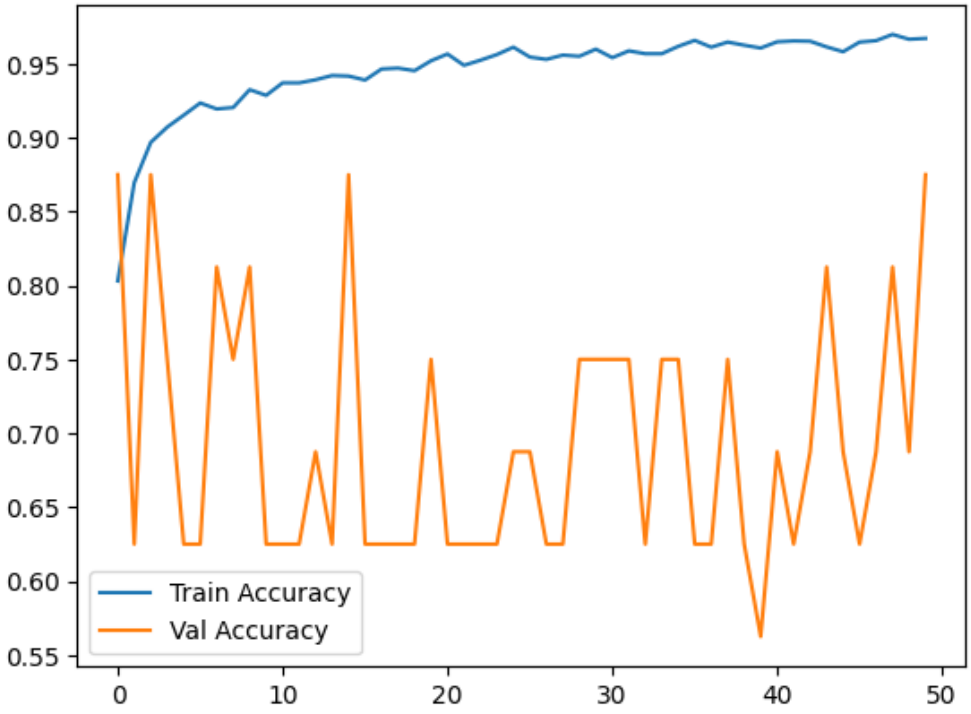

Fig. 6: Accuracy Curve of CNN

The CNN accuracy curve shows a steady increase in training accuracy, indicating effective learning and convergence. In contrast, the validation accuracy exhibits fluctuations due to the limited size of the validation dataset. Despite this variability, the model maintains reasonable generalization performance. A slight gap between training and validation accuracy suggests mild overfitting in later epochs.

6.5 Confusion Matrix Analysis

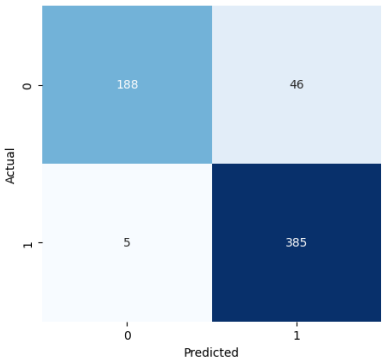

Fig. 7: CNN Confusion Matrix

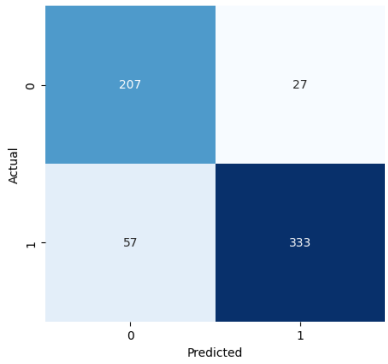

Fig. 8: ResNet50 Confusion Matrix

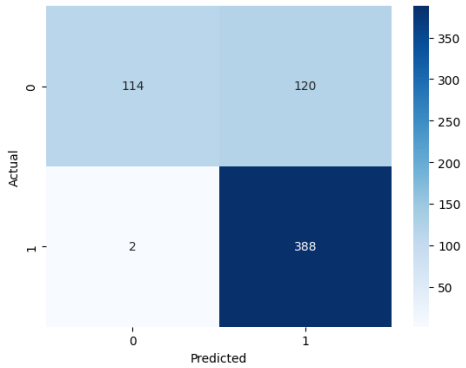

Fig. 9: MobileNetV2 Confusion Matrix

The CNN model correctly classified most Pneumonia cases with minimal false negatives. ResNet50 showed moderate errors, while MobileNetV2 misclassified a large number of Normal samples.

6.6 Comparison with Base Models

The CNN model outperformed both ResNet50 and MobileNetV2 in terms of overall accuracy and balanced classification. Although ResNet50 improved after fine-tuning, it remained slightly below CNN performance. MobileNetV2 demonstrated lower reliability due to class imbalance.

6.7 Key Observations

- CNN achieved the highest accuracy (92%)
- ResNet50 showed stable performance (87%)
- MobileNetV2 suffered from class imbalance (80%)
- Fine-tuning significantly improved ResNet performance

7. Conclusion

This study presented a comparative analysis of a custom Convolutional Neural Network (CNN) and transfer learning-based models, including ResNet50 and MobileNetV2, for pneumonia detection using chest X-ray images.

The experimental results demonstrated that the custom CNN model achieved the best overall performance, with an accuracy of **92%**, outperforming both ResNet50 and MobileNetV2 due to better adaptation to domain-specific features and dataset size constraints. The CNN model also exhibited high recall for Pneumonia cases, making it highly suitable for medical diagnosis tasks where minimizing false negatives is critical.

Although ResNet50 benefited from transfer learning and fine-tuning, its performance remained slightly lower than the CNN model, indicating that deeper architectures do not always guarantee better results, especially when applied to domain-specific datasets. MobileNetV2, while computationally efficient, showed significant bias toward the Pneumonia class and lower overall reliability.

Overall, the study highlights that a well-designed CNN can effectively learn domain-specific features and outperform complex pre-trained models under certain conditions. Future work may focus on improving performance through larger datasets, advanced architectures, and better handling of class imbalance.

8. References

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